



Intelligent Narmada Basin Monitoring & Governance (INBMG)

*An AI-Driven Framework for Ecological Conservation
and Public Safety*



AI-Powered
Safety



Remote
Sensing



eDNA Based
Monitoring



Smart
Governance



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CHALLENGES IN NARMADA BASIN MANAGEMENT

Complex Problems. Critical Impact.



01



CROWD SAFETY RISK

High-density gatherings at ghats increase risk of stampedes and drowning.



02



UNPREDICTABLE RIVER NAVIGATION

Dynamic water flow and shifting sandbars make navigation unsafe.



03



ENVIRONMENTAL DEGRADATION

Manual and slow monitoring leads to biodiversity loss and pollution.



Lack of real-time data, prediction, and integrated systems leads to reactive decisions and high risks.





LIMITATIONS OF EXISTING SYSTEMS

Why traditional river management approaches fall short



1. REACTIVE APPROACH

Actions are taken only after incidents occur, not before.



2. LACK OF REAL-TIME DATA

No continuous monitoring or live insights for decision-making.



3. FRAGMENTED SYSTEMS

Data and tools are scattered, leading to poor coordination.



Disconnected data, delayed actions, and poor coordination result in **higher risks and greater losses.**





Introducing **INBMG:** A Smart River Intelligence System

From Reactive Response to **Predictive Intelligence**



AI-Based Safety Monitoring

Real-time crowd analytics
and risk prediction



Remote Sensing & Data Analytics

Satellite monitoring and
AI-powered insights



Intelligent Navigation System

Safe routes, depth prediction
and hazard alerts



River Health & eDNA Monitoring

Real-time water quality
and biodiversity tracking



Predict Risks



Protect Lives



Preserve Ecosystems

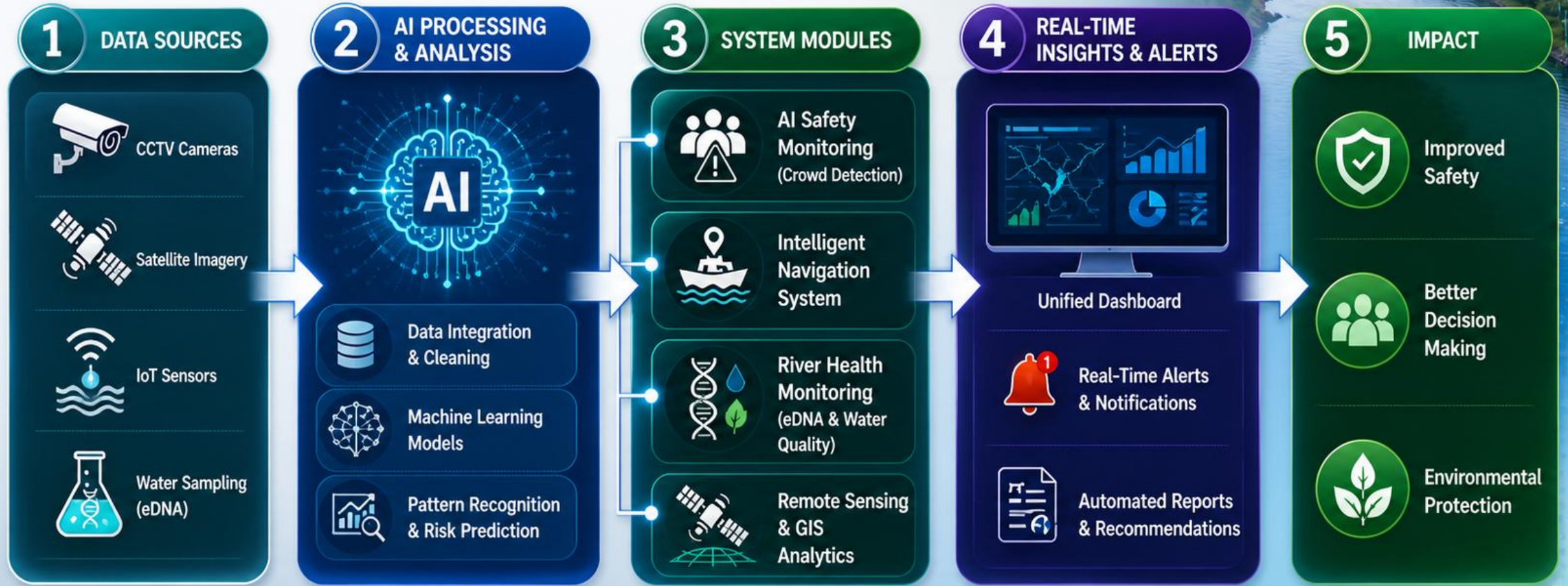


Enable Data-Driven
Governance



INBMG SYSTEM WORKFLOW

From Data to Decisions. **Intelligence in Action.**



An Integrated AI Platform for a Safer, Smarter & Sustainable **Narmada Basin**





AI-BASED SAFETY MONITORING SYSTEM

Real-Time Crowd Intelligence for Public Safety

CROWD DENSITY

- High Risk
- Elevated
- Moderate
- Low

LIVE STATUS

MONITORING



AI MODELS

Continuously analyzing crowd behavior in real-time



1. Real-Time Crowd Detection

AI models monitor crowd density continuously.



2. Risk Prediction & Alerts

Detects dangerous crowd build-up before incidents occur.



3. Emergency Response System

Automated alerts and rapid response actions (e.g., drone-assisted rescue).



Prevent
Accidents



Save
Lives



Faster
Response



Informed
Decisions



INTELLIGENT RIVER NAVIGATION SYSTEM

AI-POWERED SAFE NAVIGATION FOR DYNAMIC RIVER CONDITIONS

REAL-TIME DATA

 Water Flow
Moderate

 Water Depth
4.2 m

 Obstacles
Detected

 Weather
Clear

NAVIGATION DASHBOARD



SPEED
18
km/h

ETA
00:18:45

DESTINATION
Narmada Ghat

SAFE ROUTE


SHALLOW
DEPTH

STRONG
CURRENT 


ROCKY AREA



AI ENSURES SAFE, SMART & EFFICIENT NAVIGATION



1. SAFE ROUTE PREDICTION

AI suggests the optimal and safest navigation path in real-time.



2. RISK DETECTION & ALERTS

Identifies dangerous zones, shallow depth, and collision risks.



3. VOICE GUIDANCE SYSTEM

Provides real-time voice alerts and guidance in local languages.



PREVENTS ACCIDENTS
Reduces risk and saves lives



ENSURES PASSENGER SAFETY
Safe journeys for all



IMPROVES EFFICIENCY
Optimized routes, saved time



**SUPPORTS SUSTAINABLE
RIVER TRANSPORT**
Eco-friendly and smart mobility



RIVER HEALTH & eDNA MONITORING

AI-Powered Real-Time Biodiversity & Water Quality Assessment

NEXT-GEN SCIENCE.
SMARTER RIVERS.
HEALTHIER FUTURE.



HOW eDNA MONITORING WORKS

1



WATER SAMPLE COLLECTION

Collect water from multiple locations.

2



DNA EXTRACTION & ANALYSIS

Extract and isolate DNA fragments.

3



AI-BASED IDENTIFICATION

AI models analyze eDNA sequences.

4



DATABASE MATCHING

Match sequences with reference databases.

5



BIODIVERSITY INSIGHTS

Generate biodiversity & ecological insights.



WHAT WE MONITOR



1. WATER QUALITY MONITORING

Tracks pollution levels, pH, dissolved oxygen, and chemical indicators.



2. BIODIVERSITY DETECTION

Identifies species presence using eDNA analysis – from microbes to fish & plants.



3. ECOSYSTEM HEALTH ANALYSIS

Evaluates ecological balance, tracks changes over time, and identifies environmental risks early.

WATER QUALITY INDEX



SPECIES DETECTED



ECOSYSTEM HEALTH SCORE



Early Pollution
Detection



Protects
Biodiversity



Supports
Sustainable Decisions



Data-Driven
Environmental Action



KNOW THE RIVER. PROTECT THE FUTURE.



REMOTE SENSING & GIS ANALYTICS



Large-Scale Monitoring of River Systems Using Satellite Data

MAP LAYERS

- Water Bodies
- Flood Zones
- Vegetation
- Agriculture
- Built-up Areas
- Monitoring Points

VEGETATION

AGRICULTURAL LAND

MONITORING STATION

FLOOD ZONE

HIGH RISK AREA

REAL-TIME SATELLITE INSIGHTS

- Hi-Resolution Satellite Data
- Frequent Monitoring
- AI + GIS Integration
- Actionable Insights



1. FLOOD & WATER FLOW MONITORING

Tracks river flow patterns and predicts flood risks in real-time.



2. LAND USE & ENCROACHMENT DETECTION

Identifies changes in land use and illegal encroachment near river zones.



3. ENVIRONMENTAL CHANGE ANALYSIS

Monitors long-term ecological and geographical changes for better decision making.



Early Risk Detection



Informed Decision Making



Sustainable River Management



Protecting Communities & Ecosystems



Data-Driven Policy & Planning



FROM SATELLITE TO SOLUTION –
FOR A **SAFER, HEALTHIER RIVER BASIN.**



Unified Digital Platform

Real-Time Insights, Centralized Control, Smarter Decisions



Integrated Intelligence



Real-Time Data Fusion



Actionable Insights

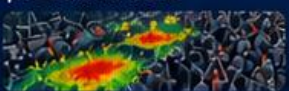


Better Outcomes



SAFETY MONITORING

Crowd detection, risk prediction & alerts



NAVIGATION SYSTEM

Safe routes, risk alerts & voice guidance



RIVER HEALTH (eDNA)

Biodiversity, water quality & ecosystem analysis



REMOTE SENSING & GIS

Satellite monitoring, flood mapping & land analysis



RIVER INTELLIGENCE SYSTEM

LIVE



12



10:30 AM
May 23, 2025

Dashboard

Map View

Alerts

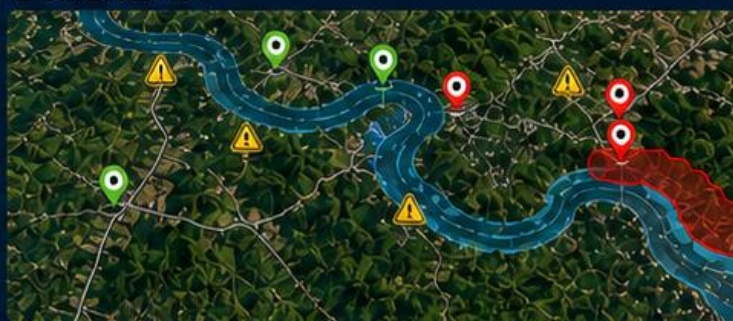
Analytics

Reports

Devices

Settings

LIVE RIVER OVERVIEW



SYSTEM STATUS



Overall Status
SAFE



Active Alerts
12



Monitoring Sites
48



Connected Devices
125



WATER QUALITY INDEX
78
Good



CROWD DENSITY
Moderate



RIVER FLOW RATE
2.35 m³/s
Normal

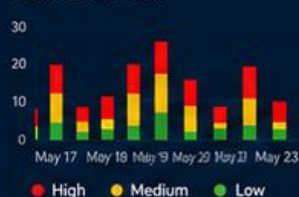


WEATHER
25°C
Clear

RIVER FLOW TREND (7 DAYS)



ALERTS OVER TIME



RECENT ALERTS

	High Crowd Density Har Ki Pauri Ghat	Just now
	High Risk Navigation Zone Near Ram Jhula	15 min ago
	Water Quality Warning Downstream Sector	30 min ago

[View All Alerts](#) →



1. REAL-TIME MONITORING

Continuous data updates from all modules and field devices.



2. SMART ALERTS & NOTIFICATIONS

Instant alerts for risks, emergencies and critical events.



3. DECISION SUPPORT SYSTEM

Data-driven insights and analytics for authorities and stakeholders.



One Platform



All Modules Integrated



Real-Time Decisions



Better Coordination



Safer Rivers,
Better Lives



One Platform. Complete Intelligence.
For a **Safer, Smarter, Sustainable Future.**



TECHNICAL STACK

Technologies Powering Intelligent River Management



AI-Powered



Data-Driven



Scalable



Secure



Real-Time

1. AI & MACHINE LEARNING



YOLOv8

Crowd Detection



CNN Models

Image Analysis



Random Forest

Classification

2. DATA & SENSORS



CCTV Cameras

Video Data Capture



IoT Sensors

Water Quality, Flow, Level, pH, DO, Turbidity



eDNA Sampling

Biodiversity Monitoring

3. NAVIGATION & ALGORITHMS



A* Algorithm

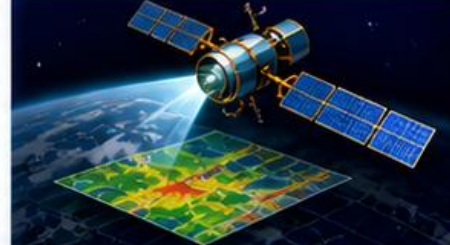
Pathfinding & Route Optimization



Risk Prediction Models

Danger Zone & Collision Risk Analysis

4. REMOTE SENSING



Satellite Data

Sentinel / LISS-IV (Multi-spectral Data)



GIS Mapping Tools

QQIS, Google Earth Engine, ArcGIS

5. PLATFORM & INTERFACE



Web Dashboard

Real-time Monitoring & Control



Data Visualization

Charts, Graphs, Heatmaps, Reports

6. INFRASTRUCTURE & BACKEND



Python



TensorFlow



PyTorch



OpenCV



PostgreSQL



MongoDB

PostgreSQL



Firebase



Docker



AWS Cloud



REST APIs



A Modern, Secure & Scalable Stack for **Smarter** and **Sustainable** River Management



IMPLEMENTATION PLAN

—●— Phased Deployment for Scalable and Real-World Impact —●—



OUR GOAL
Smarter monitoring.
Faster response.
Healthier rivers.

1

PHASE 1

Data Collection & Infrastructure



Install sensors and CCTV systems



Set up data pipelines and storage



Initial data collection and baseline analysis

2

PHASE 2

AI Integration & System Development



Train and optimize AI models



Develop navigation and monitoring systems



Integrate all modules into a unified framework

3

PHASE 3

Deployment & Monitoring



Launch unified digital platform



Real-time monitoring, alerts & notifications



Continuous improvement and system upgrades

KEY PRINCIPLES



Scalable
Architecture



Secure &
Reliable



Stakeholder
Collaboration



Sustainable
Solutions



Continuous
Improvement



**STEP BY STEP,
TOWARDS CLEANER,
SAFER, SMARTER RIVERS.**



IMPACT & OUTCOMES

Transforming River Management Through Intelligent Systems



OUR GOAL

Smarter monitoring.
Faster response.
Healthier rivers.



AI + DATA + TECHNOLOGY = SAFER, HEALTHIER, SUSTAINABLE RIVERS

1



IMPROVED PUBLIC SAFETY

Reduces accidents through real-time monitoring and alerts.



- ✓ Real-time crowd & risk detection
- ✓ Instant alerts & early warnings
- ✓ Faster emergency response

2



BETTER DECISION MAKING

Provides data-driven insights for authorities.



- ✓ Real-time dashboards & analytics
- ✓ Actionable insights for planning
- ✓ Smarter resource allocation

3



ENVIRONMENTAL PROTECTION

Supports biodiversity conservation and pollution control.



- ✓ Water quality & biodiversity monitoring
- ✓ Pollution detection & control
- ✓ Long-term ecosystem balance

4



SMART GOVERNANCE

Enables efficient and coordinated management.



- ✓ Integrated & unified platform
- ✓ Multi-agency collaboration
- ✓ Transparent & accountable governance



Safer Communities
Fewer accidents,
more lives saved



Cleaner Rivers
Reduced pollution,
better water quality



Sustainable Future
Healthy ecosystems,
long-term balance



Empowered Society
Awareness, participation
and collaboration



STRONGER RIVERS.
STRONGER TOMORROW.



INTELLIGENCE



ECOLOGY



GOVERNANCE

Reimagining River Management with AI

An Integrated System for
Safety, Sustainability, and Smart Governance



AI



Data



Ecology

=

Smarter, Safer,
Sustainable Rivers



PROTECTING LIVES

Real-time monitoring
and early alerts



PRESERVING NATURE

Cleaner rivers,
healthier ecosystems



ENABLING SMART DECISIONS

Data-driven insights
for better outcomes



CONNECTED SYSTEMS

All modules integrated
on one platform



FASTER RESPONSE BETTER IMPACT

Timely actions,
stronger results



TOGETHER FOR A BETTER FUTURE

Communities, authorities
and technology united

Every Drop Counts. Every Decision Matters.

FOR SAFER RIVERS. FOR STRONGER COMMUNITIES. FOR A SUSTAINABLE TOMORROW.



Real-Time
Monitoring



Smart Alerts
& Notifications



Advanced Analytics
& AI Models



Unified Platform
& Integration



Secure, Scalable
& Future Ready



Thank You

— Questions & Discussion —



Safer Rivers
Stronger Communities



Sustainable
Ecosystems



Smarter Decisions
Better Outcomes



Integrated Systems
Stronger Governance

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